IGCSE 0607 Practice Questions — E10.8 — Combined Questions & Markscheme

Cumulative frequency diagrams

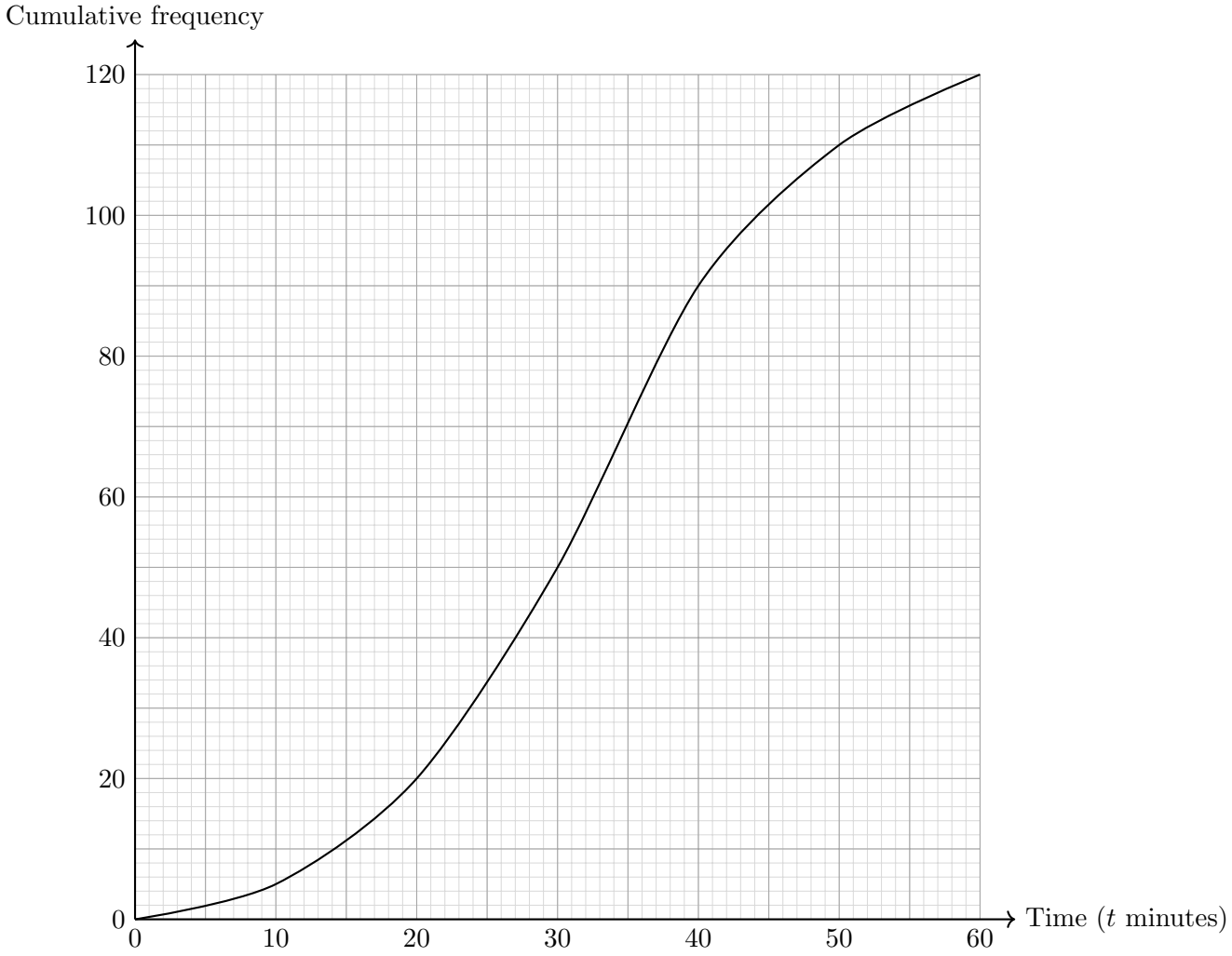
Total Questions: 9 — **Total Marks:** 68 Marks

Table of Contents & Quick Navigation

- **E10.8** Cumulative frequency diagrams
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

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1 The cumulative frequency diagram shows the times taken, t minutes, by 120 students to complete a test.



(a) Use the diagram to estimate the median time.

..... minutes **[1]**

(b) Use the diagram to estimate the interquartile range.

..... minutes **[2]**

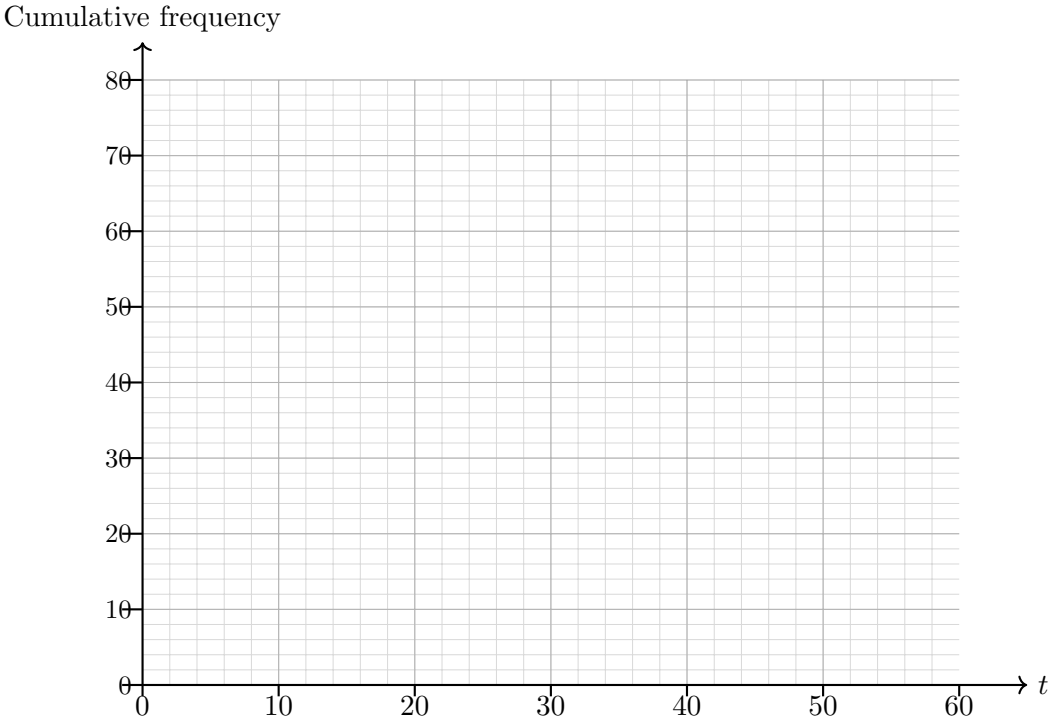
(c) Students who take longer than 45 minutes are given a penalty.
Estimate the number of students who are given a penalty.

..... **[2]**

2 The table shows the cumulative frequencies for the times, t minutes, taken by 80 students to complete a puzzle.

Time (t minutes)	$t \leq 10$	$t \leq 20$	$t \leq 30$	$t \leq 40$	$t \leq 50$	$t \leq 60$
Cumulative frequency	6	22	48	68	78	80

(a) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(b) Use your diagram to estimate the median time.

..... minutes **[2]**

3 A company records the salaries, s thousand dollars, of its 200 employees.
The frequency table shows the results.

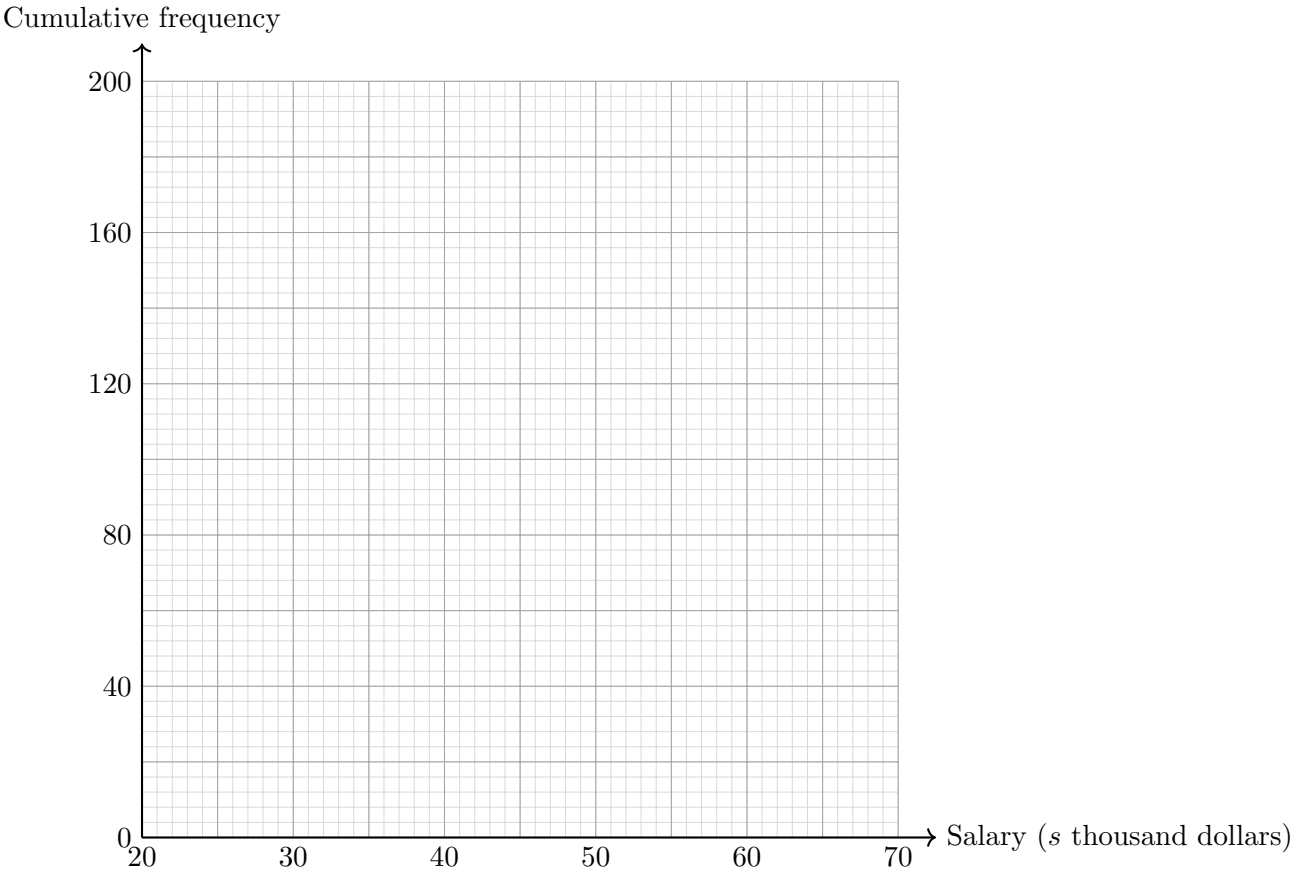
Salary (s thousand dollars)	$20 < s \leq 30$	$30 < s \leq 40$	$40 < s \leq 50$	$50 < s \leq 60$	$60 < s \leq 70$
Frequency	24	66	70	28	12

(a) Complete the cumulative frequency table.

Salary (s thousand dollars)	$s \leq 30$	$s \leq 40$	$s \leq 50$	$s \leq 60$	$s \leq 70$
Cumulative frequency	24				

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.

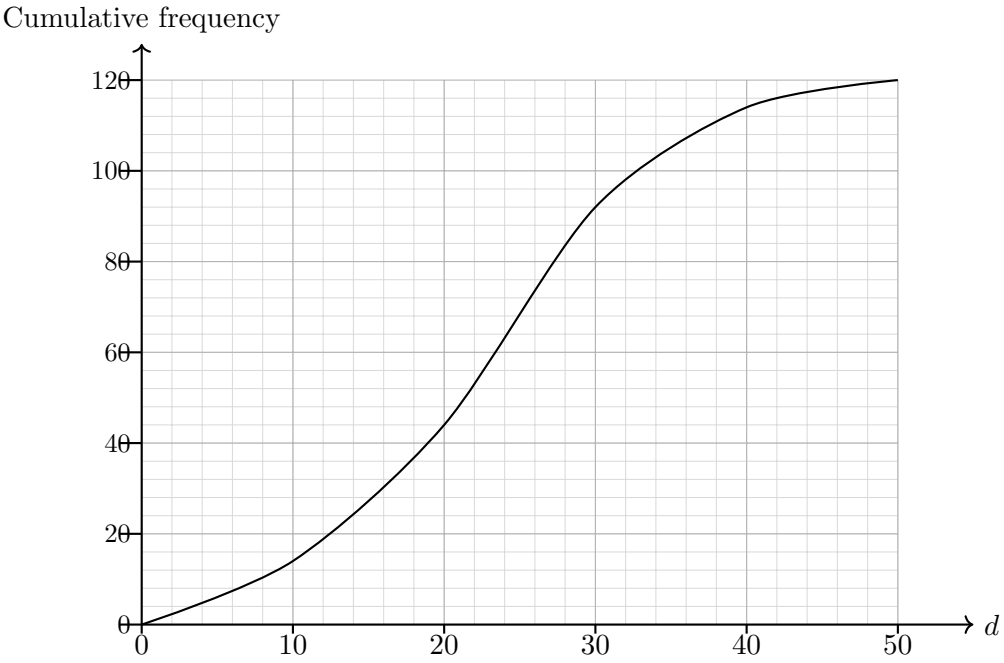


[3]

(c) Use your diagram to estimate the 70th percentile of the salaries.

..... thousand **[2]**

4 The cumulative frequency diagram shows information about the distances, d km, travelled by 120 people to work.



(a) Use the diagram to estimate the median distance.

..... km **[2]**

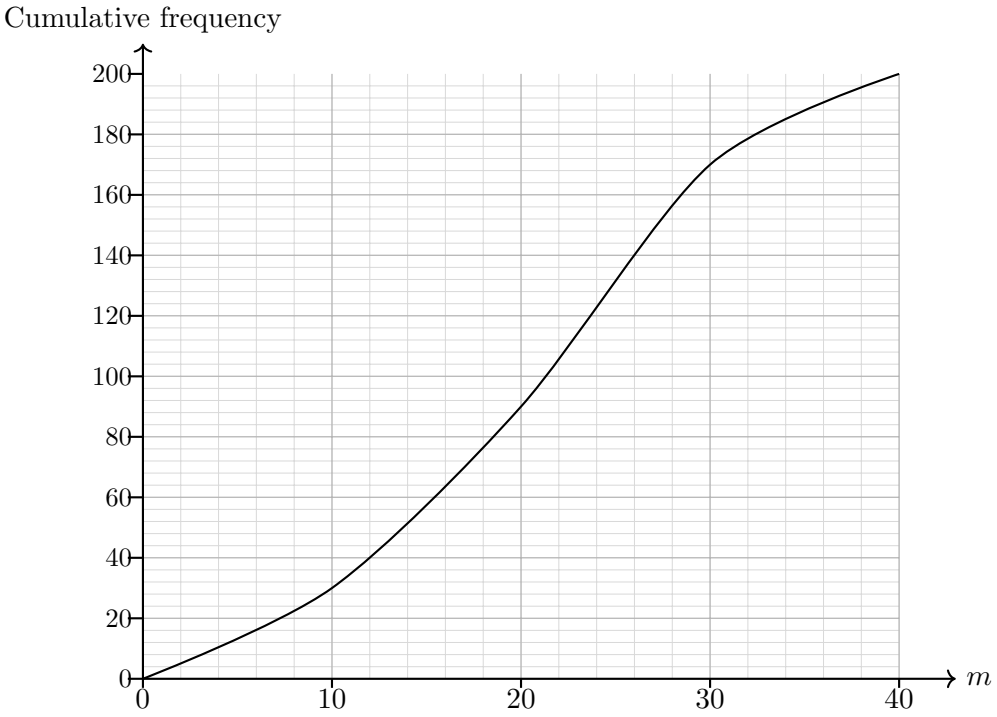
(b) Use the diagram to estimate the interquartile range.

..... km **[3]**

(c) Estimate the number of people who travelled more than 35 km to work.

..... **[3]**

5 The cumulative frequency diagram shows information about the masses, m grams, of 200 parcels.



(a) Find an estimate for the 80th percentile.

..... grams **[2]**

(b) Use the cumulative frequency diagram to complete the frequency table.

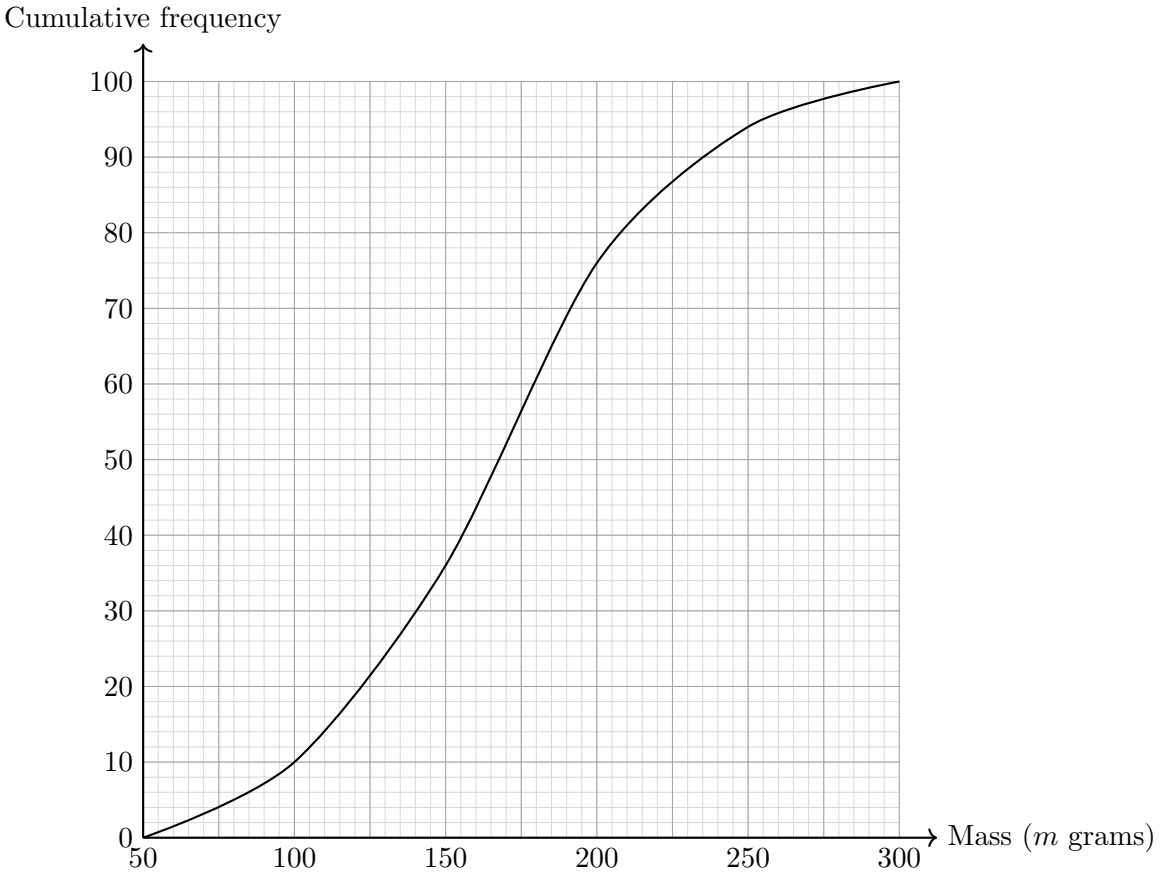
Mass (m grams)	$0 < m \leq 10$	$10 < m \leq 20$	$20 < m \leq 30$	$30 < m \leq 40$
Frequency				

[2]

(c) Calculate an estimate of the mean mass.

..... grams [4]

6 The cumulative frequency curve shows information about the masses, m grams, of 100 potatoes.



(a) Use the curve to estimate the number of potatoes with a mass greater than 220 grams.

..... **[3]**

(b) The curve was drawn using grouped data.
Find the values of p, q, and r.

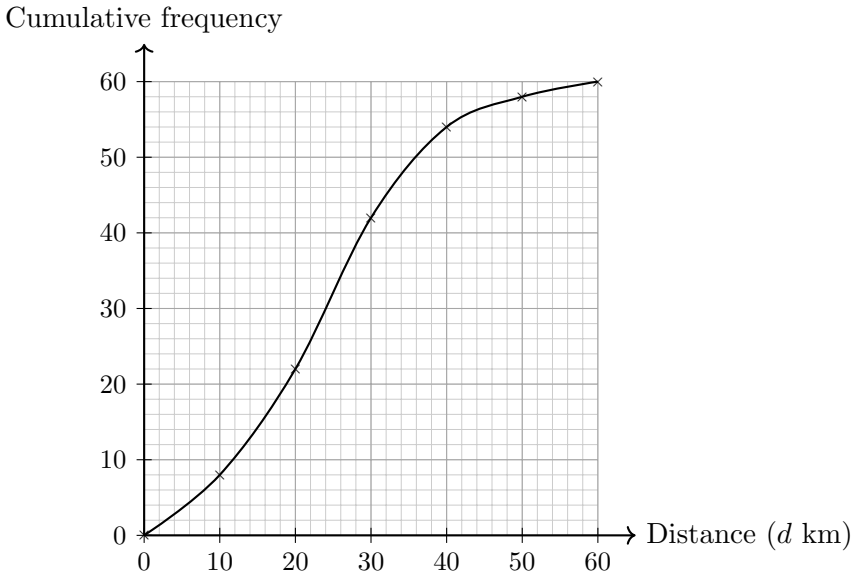
Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	p	q	40	r	6

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

(c) Calculate an estimate of the mean mass of the potatoes.

..... g **[3]**

7 The cumulative frequency diagram shows information about the distance, d km, travelled by 60 people to work.



(a) Use the diagram to find an estimate for:

(i) the median,

..... km **[1]**

(ii) the interquartile range.

..... km **[2]**

(b) Find the number of people who travelled more than 45 km to work.

..... **[3]**

8 A table shows information about the masses of 120 apples.

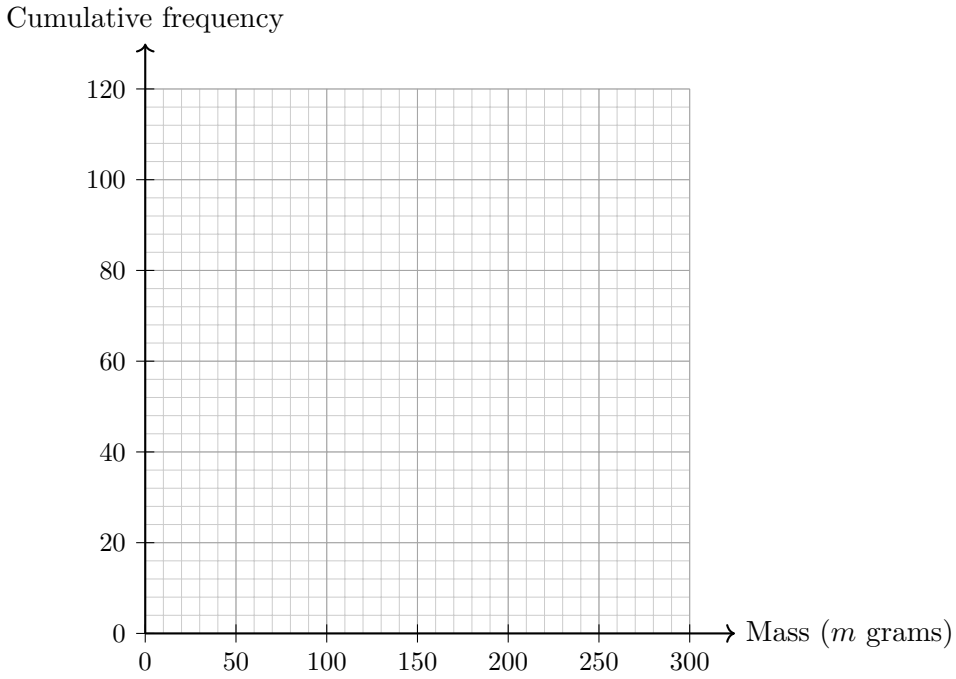
Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	12	34	46	20	8

(a) Complete the cumulative frequency table.

Mass (m grams)	$m \leq 100$	$m \leq 150$	$m \leq 200$	$m \leq 250$	$m \leq 300$
Cumulative frequency	12				120

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

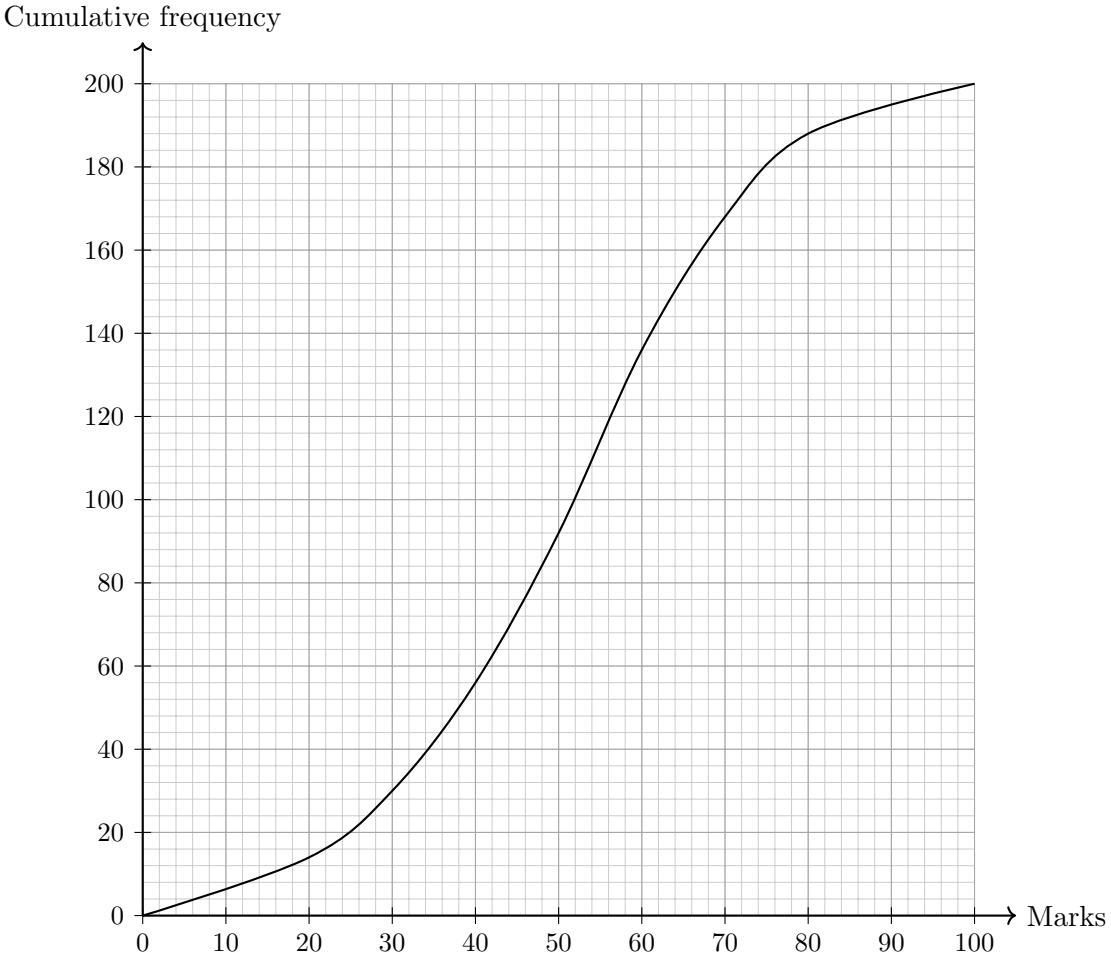
(c) Use your diagram to estimate the 60th percentile.

..... g **[2]**

(d) Apples with a mass greater than 220 grams are classified as large.
Estimate the number of large apples.

..... **[3]**

9 The cumulative frequency diagram shows the distribution of marks scored by 200 candidates in an examination.



(a) Use your graph to estimate the interquartile range.

..... **[2]**

(b) 15% of the candidates score less than m marks.
Find the value of m.

$m =$ ..... **[2]**

(c) Candidates scoring more than P marks are awarded a Distinction.
Given that 30 candidates are awarded a Distinction, find the value of P.

$P =$ ..... **[2]**

(d) Candidates who score less than 45 marks have to take a supplementary test.
Calculate the percentage of candidates who do NOT have to take this test.

..... % **[4]**

1 **(a)** 32 to 34 **[B1]**

(b) Reading Upper Quartile as 40 and Lower Quartile as 23 to 24 **[M1]**

16 to 17.5 **[A1]**

(c) $120 - 100$ (or 120 – their reading at $t = 45$) **[M1]**

18 to 22 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** 6 points plotted correctly at the upper boundaries **[P2]**

(Allow P1 for 4 or 5 points plotted correctly)

A smooth curve drawn passing through all their points **[C1]**

(b) Reading from cumulative frequency 40 **[M1]**

$26 \leq$ answer ≤ 28 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $90, 160, 188, 200$ **[B2]**

(B1 for 2 or 3 correct values)

(b) 5 points plotted correctly from their table **[B2]**

(B1 for 3 or 4 points plotted correctly)

Smooth curve drawn through their points **[B1]**

(c) Reading from curve at cumulative frequency = 140 **[M1]**

46 to 48 **[A1]**

[Back to Question] *[Total Marks: 7]*

4 **(a)** Reading from cumulative frequency 60 **[M1]**

$22 \leq$ answer ≤ 24 **[A1]**

(b) Attempt to read from cumulative frequencies 90 (UQ) and 30 (LQ) **[M1]**

Lower quartile ≈ 15.5 and Upper quartile ≈ 29.5 seen **[M1]**

$12 \leq$ answer ≤ 16 **[A1]**

(c) Attempt to read at $d = 35$ (approx 105) **[M1]**

120 – their reading **[M1]**

$13 \leq$ answer ≤ 17 **[A1]**

[Back to Question] *[Total Marks: 8]*

5 **(a)** $0.8 \times 200 = 160$ **[M1]**

$28 \leq$ answer ≤ 29 **[A1]**

(b) Frequencies 30, 60, 80, 30 correct **[B2]**

(Allow B1 for 2 or 3 correct frequencies)

(c) Using midpoints 5, 15, 25, 35 **[M1]**

$\Sigma fx = 30 \times 5 + 60 \times 15 + 80 \times 25 + 30 \times 35$ (using their frequencies) **[M1]**

their $4100 \div 200$ **[M1]**

20.5 **[A1]**

[Back to Question] *[Total Marks: 8]*

6 **(a)** Reading from curve at $m = 220$: 81 to 85 **[M1]**

$100 -$ their reading **[M1]**

15 to 19 **[A1]**

(b) $p = 10$ **[B1]**

$q = 26$ **[B1]**

$r = 18$ **[B1]**

(c) Use of correct midpoints: $75, 125, 175, 225, 275$ **[M1]**

$\sum f \times m = 10 \times 75 + 26 \times 125 + 40 \times 175 + 18 \times 225 + 6 \times 275 = 16700$ **[M1]**

$16700 \div 100 = 167$ **[A1]**

[Back to Question] *[Total Marks: 9]*

7 **(a)** **(i)** 24 **[B1]**

(ii) Reading Lower Quartile = 15 and Upper Quartile = 32.5 (Accept 32 to 33) **[M1]**

17.5 (Accept 17 to 18) **[A1]**

(b) Reading Cumulative Frequency at 45 km = 56 (Accept 55 to 57) **[M1]**

$60 - 56$ **[M1]**

4 (Accept 3 to 5) **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** 46, 92, 112 **[B2]**

(Allow [B1] for 2 correct values)

(b) Points correctly positioned horizontally and vertically starting at $(50, 0)$ **[B1]**

All 5 points plotted correctly at class upper boundaries (Tolerance $\pm$ half small square) **[B1]**

Smooth curve drawn through all 6 points **[B1]**

(c) Finding target Cumulative Frequency = 60% of 120 = 72 **[M1]**

178 (Accept 176 to 180) **[A1]**

(d) Reading Cumulative Frequency at $m = 220$ is 100 **[M1]**

$120 - 100$ **[M1]**

20 (Accept 18 to 22) **[A1]**

[Back to Question] *[Total Marks: 10]*

9 **(a)** Reading Lower Quartile = 38 (Accept 37 to 39) and Upper Quartile = 64 (Accept 63 to 66) **[M1]**

26 (Accept 24 to 29) **[A1]**

(b) Finding Cumulative Frequency = 15% of 200 = 30 **[M1]**

30 (Accept 29 to 31) **[A1]**

(c) Finding target Cumulative Frequency = $200 - 30 = 170$ **[M1]**

71 (Accept 70 to 72) **[A1]**

(d) Reading Cumulative Frequency at 45 marks = 74 (Accept 72 to 76) **[M1]**

Number of candidates NOT taking test = $200 - 74 = 126$ **[M1]**

$\frac{126}{200} \times 100\%$ **[M1]**

63 (Accept 62 to 64) **[A1]**

[Back to Question] *[Total Marks: 10]*